

117. Reconciliation Sublayer (RS) and Media Independent Interface for 200 Gb/s and 400 Gb/s operation (200GMII and 400GMII)

117.1 Overview

This clause defines the characteristics of the Reconciliation Sublayer (RS) and the Media Independent Interface between Ethernet media access controllers and various PHYs. Figure 117–1 shows the relationship of the RS and Media Independent Interface to the ISO/IEC OSI reference model.

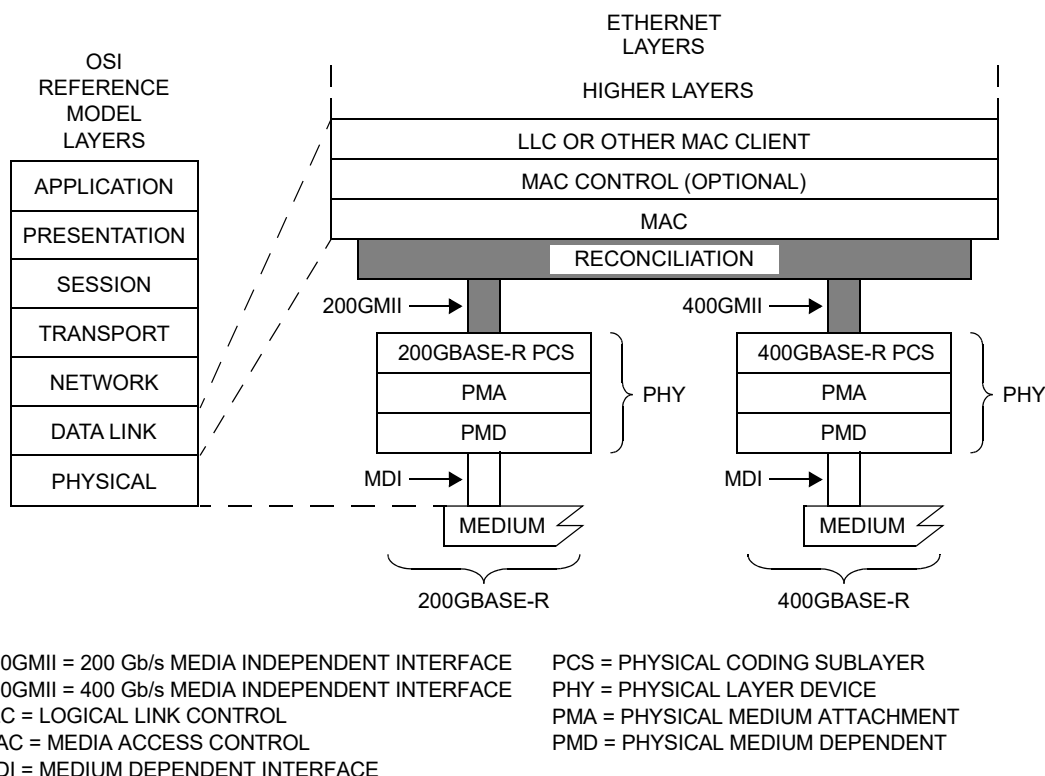


Figure 117–1—RS and MII relationship to the ISO/IEC Open Systems Interconnection (OSI) reference model and IEEE 802.3 Ethernet model

The 200GMII and the 400GMII are optional logical interfaces between the MAC sublayer and the Physical Layer device (PHY). The 200GXS/400GXS sublayer in conjunction with the 200GAUI-n/400GAUI-n interface may be used to optionally extend the 200GMII/400GMII.

The RS adapts the bit serial protocols of the MAC to the parallel format of the PCS service interface. Though the 200GMII and 400GMII are optional interfaces, they are used in this standard as a basis for specification. The Physical Coding Sublayer (PCS) is specified to the 200GMII/400GMII, so if not implemented, a conforming implementation shall behave functionally as if the RS and 200GMII/400GMII were implemented.

The 200GMII/400GMII have the following characteristics:

- The 200GMII supports a speed of 200 Gb/s.
- The 400GMII supports a speed of 400 Gb/s.

- c) Data and delimiters are synchronous to a clock reference.
- d) They provide independent 64-bit wide transmit and receive data paths.
- e) They support full duplex operation only.

117.1.1 Summary of major concepts

The following are the major concepts of the 200GMII/400GMII:

- a) The 200GMII/400GMII are functionally similar to other media independent interfaces that have been defined for lower speeds, as they all define an interface allowing independent development of MAC and PHY logic.
- b) The RS converts between the MAC serial data stream and the parallel data paths of the 200GMII/400GMII.
- c) The RS maps the signal set provided at the 200GMII/400GMII to the PLS service primitives provided at the MAC.
- d) Each direction of data transfer is independent and serviced by data, control, and clock signals.
- e) The RS generates continuous data or control characters on the transmit path and expects continuous data or control characters on the receive path.
- f) The RS participates in link fault detection and reporting by monitoring the receive path for status reports that indicate an unreliable link, and generating status reports on the transmit path to report detected link faults to the DTE on the remote end of the connecting link.
- g) The 200GMII/400GMII may also support Low Power Idle (LPI) signaling for PHY types supporting Energy Efficient Ethernet (EEE) (see Clause 78).
- h) The 200GMII/400GMII can be extended through the use of a pair of 200GXS/400GXS sublayers (DTE XS and PHY XS) with a 200GAUI-n/400GAUI-n between them.

117.1.2 Application

This clause applies to the interface between the MAC and PHY. This logical interface is used to provide media independence so that an identical media access controller may be used with supported PHY types.

117.1.3 Rate of operation

The 200GMII is specified to support 200 Gb/s operation. The 400GMII is specified to support 400 Gb/s operation.

117.1.4 Delay constraints

The maximum cumulative MAC Control, MAC, and RS delay (sum of transmit and receive delays at one end of the link) shall meet the values specified in Table 117–1. A description of overall system delay constraints and the definitions for bit times and pause_quanta can be found in 116.4 and its references.

Table 117–1—Delay constraints

Sublayer	Maximum (bit time)	Maximum (pause_quanta)	Maximum (ns)
200 Gb/s MAC, RS, and MAC Control	49 152	96	245.76
400 Gb/s MAC, RS, and MAC Control	98 304	192	245.76

117.1.5 Allocation of functions

The allocation of functions at the 200GMII/400GMII balances the need for media independence with interface simplicity. The 200GMII/400GMII provides media independence by separating the Data Link and Physical Layers of the OSI seven-layer reference model.

117.1.6 200GMII/400GMII structure

The 200GMII/400GMII structure is identical to the CGMII structure specified in 81.1.6.

117.1.7 Mapping of 200GMII/400GMII signals to PLS service primitives

The Reconciliation Sublayer (RS) shall map the signals provided at the 200GMII/400GMII to the PLS service primitives defined in Clause 6. The PLS service primitives provided by the RS and described here behave in exactly the same manner as defined in Clause 6. Full duplex operation only is implemented at either 200 Gb/s or 400 Gb/s, therefore, PLS service primitives supporting CSMA/CD operation are not mapped through the RS to the 200GMII/400GMII. This behavior and restrictions are the same as described in 22.7, with the details of the signaling described in 117.3. LPI_REQUEST shall not be set to ASSERT unless the attached link has been operational for at least one second (i.e., link_status = OK, according to the underlying PCS/PMA).

EEE capability requires the use of the MAC defined in Annex 4A for simplified full duplex operation (with carrier sense deferral). This provides full duplex operation but uses the carrier sense signal to defer transmission until the system returns from its low power state.

Mappings for the following primitives are defined for 200 Gb/s and 400 Gb/s operation:

- PLS_DATA.request
- PLS_DATA.indication
- PLS_CARRIER.indication
- PLS_SIGNAL.indication
- PLS_DATA_VALID.indication

The RS maps all primitives in an identical manner as the CGMII does and as specified in 81.1.7.

117.2 200GMII/400GMII data stream

The 200GMII/400GMII data stream has the same characteristics as the CGMII data stream described in 81.2.

117.3 200GMII/400GMII functional specifications

The 200GMII/400GMII functions identically to the CGMII specified in 81.3.

117.4 LPI Assertion and Detection

LPI assertion and detection function identically to the CGMII specified in 81.4, with the single exception that the PMA stop signaling described in 81.4.4 is not applicable.

117.5 Protocol implementation conformance statement (PICS) proforma for Clause 117, Reconciliation Sublayer (RS) and Media Independent Interface for 200 Gb/s and 400 Gb/s operation (200GMII and 400GMII)²⁰⁴

117.5.1 Introduction

The supplier of a protocol implementation that is claimed to conform to Clause 117, Reconciliation Sublayer (RS) and Media Independent Interface for 200 Gb/s and 400 Gb/s operation (200GMII and 400GMII), shall complete the following protocol implementation conformance statement (PICS) proforma.

A detailed description of the symbols used in the PICS proforma, along with instructions for completing the PICS proforma, can be found in Clause 21.

117.5.2 Identification

117.5.2.1 Implementation identification

Supplier ¹	
Contact point for inquiries about the PICS ¹	
Implementation Name(s) and Version(s) ^{1,3}	
Other information necessary for full identification—e.g., name(s) and version(s) for machines and/or operating systems; System Name(s) ²	
NOTE 1—Required for all implementations. NOTE 2—May be completed as appropriate in meeting the requirements for the identification. NOTE 3—The terms Name and Version should be interpreted appropriately to correspond with a supplier's terminology (e.g., Type, Series, Model).	

117.5.2.2 Protocol summary

Identification of protocol standard	IEEE Std 802.3-2022, Clause 117, Reconciliation Sublayer (RS) and Media Independent Interface for 200 Gb/s and 400 Gb/s operation (200GMII and 400GMII)
Identification of amendments and corrigenda to this PICS proforma that have been completed as part of this PICS	
Have any Exception items been required? No [] Yes [] (See Clause 21; the answer Yes means that the implementation does not conform to IEEE Std 802.3-2022.)	
Date of Statement	

²⁰⁴Copyright release for PICS proformas: Users of this standard may freely reproduce the PICS proforma in this subclause so that it can be used for its intended purpose and may further publish the completed PICS.

117.5.3 Major capabilities/options

Item	Feature	Subclause	Value/Comment	Status	Support
*MII	PHY support of either 200GMII or 400GMII	117.2, 117.3		O	Yes [] No []
*LPI	Implementation of LPI	117.1.7		O	Yes [] No []

117.5.4 PICS proforma tables for Reconciliation Sublayer (RS) and Media Independent Interface for 200 Gb/s and 400 Gb/s operation (200GMII and 400GMII)

117.5.4.1 General

Item	Feature	Subclause	Value/Comment	Status	Support
G1	Cumulative MAC Control, MAC and RS delay	117.1.4	Per Table 117–1	M	Yes []
G3	Lane structure	117.1.6	Per Table 81–2	M	Yes []

117.5.4.2 Mapping of PLS service primitives

Item	Feature	Subclause	Value/Comment	Status	Support
PL1	Mapping to Clause 6	117.1.7	RS implements mapping to Clause 6 PLS service primitives	MII:M	Yes [] N/A []
PL2	Mapping of PLS_DATA.requests	117.1.7	In sequence TXD<0> to TXD<63>	MII:M	Yes [] N/A []
PL3	Start control character creation	117.1.7	First octet of preamble converted to Start control character	MII:M	Yes [] N/A []
PL4	TXD and TXC generation	117.1.7	For each 64 PLS_DATA.requests	MII:M	Yes [] N/A []
PL5	Terminate control character creation	117.1.7	DATA_COMPLETE causes creation of Terminate control character in next lane in sequence	MII:M	Yes [] N/A []
PL6	Mapping RXD to PLS_DATA.indications	117.1.7	Create PLS_DATA.indications in sequence from RXD<0> to RXD<63>	MII:M	Yes [] N/A []
PL7	PLS_DATA.indication generation	117.1.7	Generate 64 PLS_DATA.indications for each RXD<63:0> until Terminate then generating 0, 8, 16, 24, 32, 40, 48, or 56	MII:M	Yes [] N/A []
PL8	Start control character conversion	117.1.7	Convert valid Start control character to preamble before generating PLS_DATA.indications	MII:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
PL9	Terminate control character	117.1.7	No PLS_DATA.indications generated	MII:M	Yes [] N/A []
PL10	PLS_DATA_VALID.indication generation	117.1.7	On change of value of DATA_VALID_STATUS	MII:M	Yes [] N/A []
PL11	DATA_VALID_STATUS	117.1.7	Value of DATA_VALID on a lane 0 Start control character preceded by eight idles, a Sequence ordered set, or a Terminate character	MII:M	Yes [] N/A []
PL12	DATA_VALID_STATUS	117.1.7	Value of DATA_NOT_VALID on any control character but Error	MII:M	Yes [] N/A []
PL13	Frame not ending with Terminate control character	117.1.7	Ensure MAC detects CRC error	MII:M	Yes [] N/A []

117.5.4.3 Data stream structure

Item	Feature	Subclause	Value/Comment	Status	Support
DS1	Frame transfer	117.2	Within 200GMII/400GMII data stream	MII:M	Yes [] N/A []
DS2	Bit mapping	117.2	Per Figure 81–4	MII:M	Yes [] N/A []
DS3	Content of <data>	117.2	Consist of data octets	MII:M	Yes [] N/A []
DS4	Recognition of <efd>	117.2	Terminate recognized in any lane	MII:M	Yes [] N/A []

117.5.4.4 200GMII/400GMII signal functional specifications

Item	Feature	Subclause	Value/Comment	Status	Support
FS1	TX_CLK active edge	117.3	TXD and TXC sampled on the rising edge of TX_CLK	MII:M	Yes [] N/A []
FS2	TX_CLK frequency	117.3	One-sixty-fourth of the MAC transmit data rate	MII:M	Yes [] N/A []
FS3	TXC assertion and de-assertion	117.3	De-asserted for data, asserted for control character	MII:M	Yes [] N/A []
FS4	TXC clock	117.3	Synchronous to TX_CLK	MII:M	Yes [] N/A []
FS5	TXD encoding	117.3	Per Table 81–3	MII:M	Yes [] N/A []
FS6	TXD clock	117.3	Synchronous to TX_CLK	MII:M	Yes [] N/A []
FS7	Start alignment	117.3	Start control character aligned to lane 0	MII:M	Yes [] N/A []

Item	Feature	Subclause	Value/Comment	Status	Support
FS8	RX_CLK active edge	117.3	RXD and RXC sampled on the rising edge of RX_CLK	MII:M	Yes [] N/A []
FS9	RX_CLK frequency	117.3	One-sixty-fourth of the MAC receive data rate	MII:M	Yes [] N/A []
FS10	Loss of receive signal	117.3	Source RX_CLK from nominal clock	MII:M	Yes [] N/A []
FS11	RXC assertion and de-assertion	117.3	De-asserted for data, asserted for control character	MII:M	Yes [] N/A []
FS12	RXC clock	117.3	Synchronous to RX_CLK	MII:M	Yes [] N/A []
FS13	RXD decoding	117.3	Per Table 81–4	MII:M	Yes [] N/A []
FS14	RXD clock	117.3	Synchronous to RX_CLK	MII:M	Yes [] N/A []
FS15	Received Error control character	117.3	RS cause MAC FrameCheckError	MII:M	Yes [] N/A []
FS16	DATA_VALID assertion	117.3	RS not assert DATA_VALID unless Start control character in lane 0	MII:M	Yes [] N/A []

117.5.4.5 Link fault signaling state diagram

Item	Feature	Subclause	Value/Comment	Status	Support
LF1	Link fault signaling state diagram	117.3	Implement per Figure 81–11	MII:M	Yes [] N/A []
LF2	link_fault = OK and MAC frames	117.3	RS services MAC frame transmission requests	MII:M	Yes [] N/A []
LF3	link_fault = OK and no MAC frames	117.3	In absence of MAC frames, RS transmits Idle control characters	MII:M	Yes [] N/A []
LF4	link_fault = Local Fault	117.3	RS transmits continuous Remote Fault Sequence ordered sets	MII:M	Yes [] N/A []
LF5	link_fault = Remote Fault	117.3	RS transmits continuous Idle control characters	MII:M	Yes [] N/A []

117.5.4.6 LPI functions

Item	Feature	Subclause	Value/Comment	Status	Support
L1	Assertion of LPI in Tx direction	117.3	As defined in Table 81–3	LPI:M	Yes [] N/A []
L2	Assertion of LPI in Rx direction	117.3	As defined in Table 81–4	LPI:M	Yes [] N/A []